Name: Quintana, Leigh Anne C.	Date: 11/6/2026
Subject/Section: BSCPE 3-3	Professor: Engr. Mon Arjay Malbog

Laboratory Activity
Data Classification and Preparation

I. Laboratory Objectives

At the end of this laboratory activity, students should be able to:

1. Load a dataset into a Python environment.
2. Examine the structure and characteristics of a dataset.
3. Identify and classify different variables.
4. Determine qualitative and quantitative variables.
5. Identify discrete and continuous variables.
6. Identify possible missing values and duplicate records.
7. Prepare a dataset for subsequent statistical and predictive analysis.

Software/Tools

- Python
- Jupyter Notebook / Google Colab
- Anaconda (optional)
- Pandas
- NumPy
- Matplotlib (optional)
- CSV dataset

II. Background

Data used in real-world analytics projects may contain different types of variables and may require preparation before analysis. Data classification helps analysts understand the

characteristics of each variable, while data preparation organizes and cleans the dataset so it can be used for statistical and predictive modeling.

In this activity, students will use Python and Pandas to inspect a dataset, classify its variables, identify common data problems, and produce a cleaned dataset.

III. Dataset

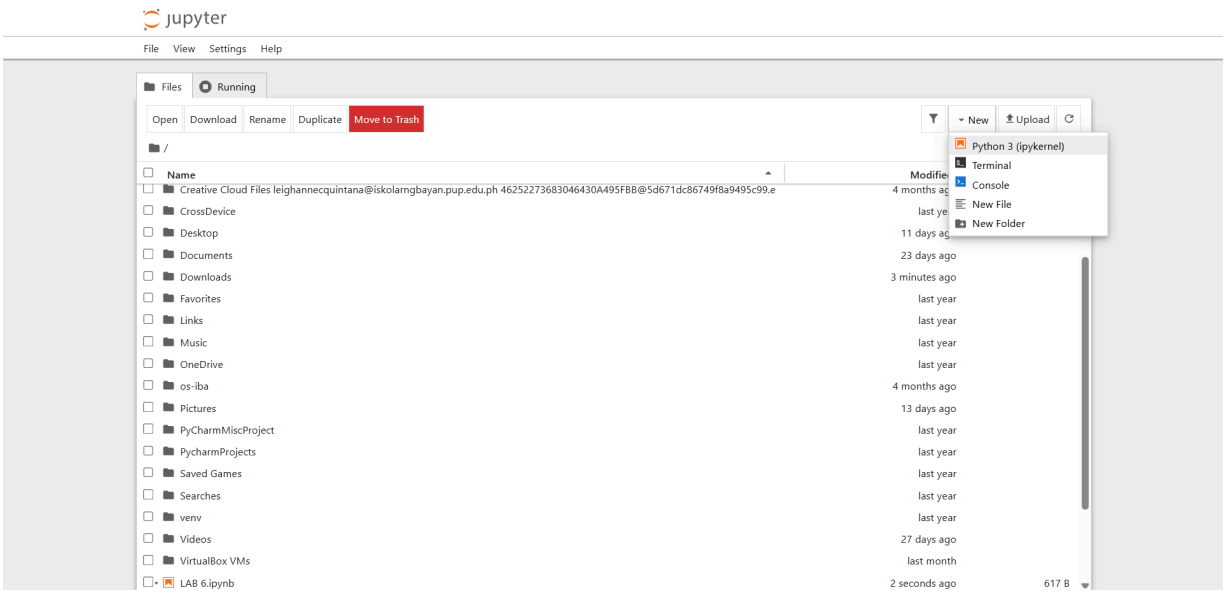
Download the dataset here: [Customer Dataset]

IV. Laboratory Procedure

Part A – Create/Load the Dataset

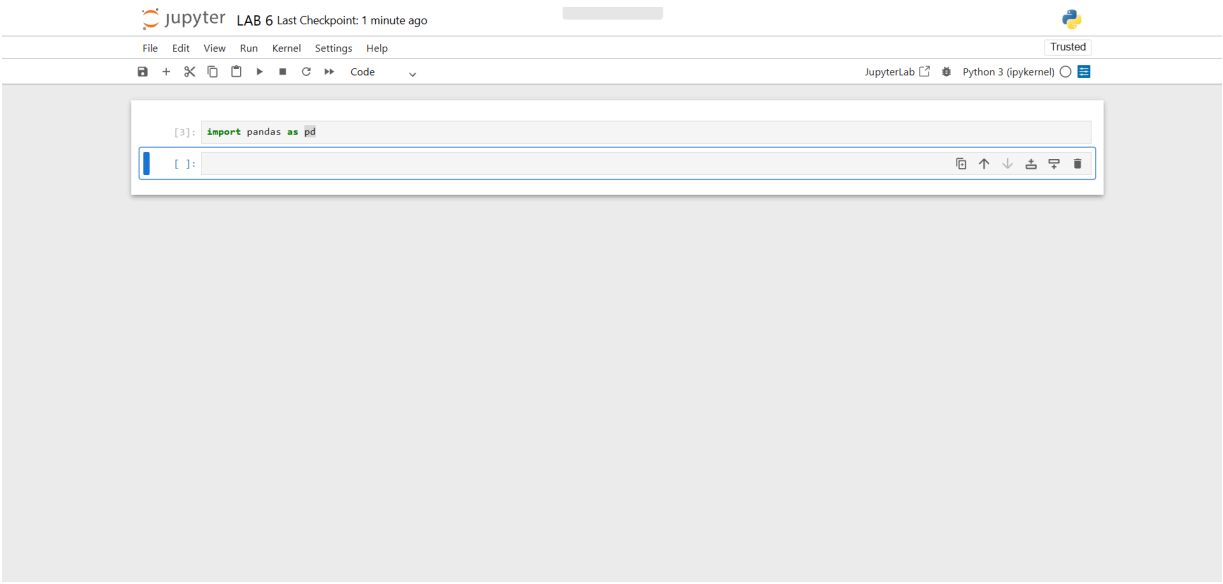
Step 1 – Open Jupyter Notebook or Google Colab

Create a new Python notebook.



Step 2 – Import Pandas

```
import pandas as pd
```



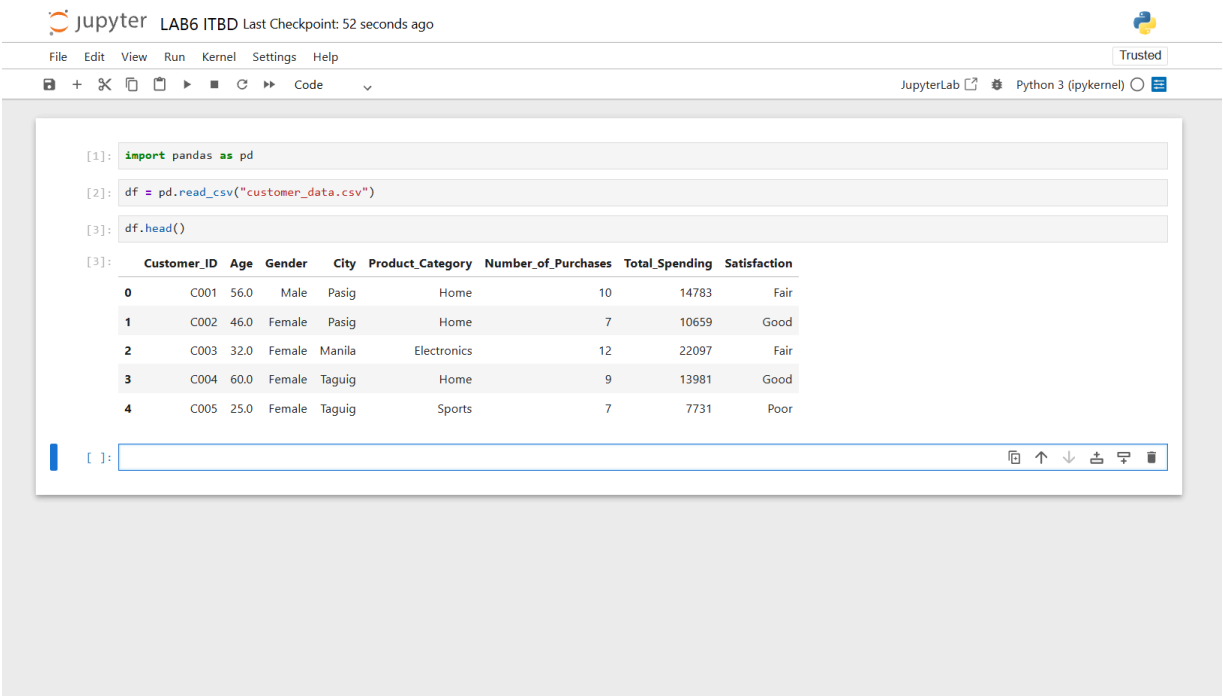
Step 3 – Load the Dataset

If using a CSV file:

```
df = pd.read_csv("customer_data.csv")
```

Display the first five records:

```
df.head()
```



Question 1

What information can you observe from the first five records?

df.head() displays the first 5 rows of your dataset. From this output, you can observe the layout of your data, the column headers (Customer_ID, Age, Gender, City, Product_Category, Number_of_Purchases, Total_Spending, Satisfaction), and sample data entries for each column.

Expected Output:

	A	B	C	D	E	F	G	H	I
	Customer_	Age	Gender	City	Product_C	Number_o	Total_Spen	Satisfaction	
	C001	56	Male	Pasig	Home	10	14783	Fair	
	C002	46	Female	Pasig	Home	7	10659	Good	
	C003	32	Female	Manila	Electronics	12	22097	Fair	
	C004	60	Female	Taguig	Home	9	13981	Good	
	C005	25	Female	Taguig	Sports	7	7731	Poor	

Part B – Inspect the Dataset

Step 4 – Determine the Number of Rows and Columns

```
df.shape
```

```
[4]: df.shape

[4]: (41, 8)

[ ]:
```

Question 2

How many:

- Rows are in the dataset? 41
- Columns are in the dataset? 8

Step 5 – Display Column Names

```
df.columns
```

```
5]: df.columns

5]: Index(['Customer_ID', 'Age', 'Gender', 'City', 'Product_Category',
       'Number_of_Purchases', 'Total_Spending', 'Satisfaction'],
      dtype='str')
```

Question 3

List all variables contained in the dataset.

Customer_ID, Age, Gender, City, Product_Category, Number_of_Purchases, Total_Spending, and Satisfaction.

Step 6 – Examine Data Types

df.dtypes

```
[6]: df.dtypes

[6]: Customer_ID               str
     Age                   float64
     Gender                    str
     City                      str
     Product_Category          str
     Number_of_Purchases     int64
     Total_Spending          int64
     Satisfaction              str
     dtype: object

[ ]:
```

Question 4

Which variables are:

- Numerical? Age, Number_of_Purchases, Total_Spending.
- Categorical/text? Customer_ID, Gender, City, Product_Category, Satisfaction.

Part C – Classify the Variables

Complete the following table based on your dataset.

Variable	Qualitative / Quantitative	Discrete / Continuous	Measurement Level
Customer_ID	Qualitative	Discrete	Nominal
Age	Quantitative	Continuous	Ratio
Gender	Qualitative	Nominal	Ratio
City	Qualitative	Nominal	Ratio

Product_Category	Qualitative	Nominal	Nominal
Number_of_Purchases	Quantitative	Discrete	Ratio
Total_Spending	Quantitative	Continuous	Ratio
Satisfaction	Qualitative	Discrete	Nominal

Question 5

Choose **three variables** and explain why you classified them that way.

Age is Quantitative because it's a numerical quantity, and Continuous because time can be broken down into fractions of years (months, days).

Gender is Qualitative because it sorts individuals into descriptive groups rather than measuring a numeric amount.

Number_of_Purchases is Quantitative and Discrete because it counts separate, whole items where fractions do not exist.

Part D – Examine Numerical Data

Step 7 – Generate Basic Information

```
df.info()
```

Observe:

- Number of entries
- Column names
- Data types
- Non-null values

```
[7]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 41 entries, 0 to 40
Data columns (total 8 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Customer_ID          41 non-null     str
 1   Age                  40 non-null     float64
 2   Gender               40 non-null     str
 3   City                 41 non-null     str
 4   Product_Category     41 non-null     str
 5   Number_of_Purchases  41 non-null     int64
 6   Total_Spending       41 non-null     int64
 7   Satisfaction         41 non-null     str
dtypes: float64(1), int64(2), str(5)
memory usage: 3.8 KB
```

```
[ ]:
```

Question 6

Are there variables with missing values? If yes, identify them.

Yes. Gender and age are both missing content

Step 8 – Generate Descriptive Information

```
df.describe()
```

Observe:

- Count
- Mean
- Standard deviation
- Minimum
- Maximum

```
memory usage: 3.8 KB
```

```
[8]: df.describe()
```

[8]:

	Age	Number_of_Purchases	Total_Spending
count	40.000000	41.000000	41.000000
mean	39.125000	8.073171	9709.268293
std	12.370742	4.107251	5625.200014
min	19.000000	1.000000	1237.000000
25%	28.750000	5.000000	6000.000000
50%	39.500000	9.000000	8908.000000
75%	47.750000	11.000000	13434.000000
max	60.000000	15.000000	25223.000000

```
[ ]:
```

Question 7

Which numerical variable has the highest average value?

Total spending has the highest average value

Part E – Identify Missing Values

Step 9 – Check Missing Values

```
df.isnull().sum()
```

This displays the number of missing values in each column.

```
[9]: df.isnull().sum()

[9]: Customer_ID           0
     Age                   1
     Gender                1
     City                  0
     Product_Category      0
     Number_of_Purchases   0
     Total_Spending        0
     Satisfaction          0
     dtype: int64

[ ]:
```

Question 8

Which variables contain missing values?

Age and gender

Question 9

How many missing values are present in each affected variable?

They each have 1 missing value

Part F – Identify Duplicate Records

Step 10 – Check for Duplicates

```
df.duplicated().sum()
```

```
[10]: df.duplicated().sum()

[10]: np.int64(1)

[ ]:
```

Question 10

Does the dataset contain duplicate records?

If duplicates exist, identify the number of duplicate rows.

There is one duplicated row.

Step 11 – Display Duplicate Records

```
df[df.duplicated()]
```

Examine the duplicate records.

```
11]: df[df.duplicated()
```

```
  Cell In[11], line 1
    df[df.duplicated()
                      ^
_IncompleteInputError: incomplete input
```

```
12]: df[df.duplicated()]
```

12]:

	Customer_ID	Age	Gender	City	Product_Category	Number_of_Purchases	Total_Spending	Satisfaction
40	C012	41.0	Male	Pasig	Clothing	9	8811	Fair

```
[ ]:
```

Part G – Check Data Consistency

Examine categorical variables:

```
df["Gender"].value_counts()
```

Or

```
df["Product_Category"].value_counts()
```

```
[13]: df["Product_Category"].value_counts()
```

```
[13]: Product_Category
      Home           10
      Grocery        10
      Sports          9
      Clothing        6
      Electronics     5
      electronics     1
      Name: count, dtype: int64
```

```
[ ]:
```

Question 13

Did you find inconsistent category values?

If yes, identify them.

Yes. Electronics and electronics are treated as 2 different entities.